

Assignment No-02 Wireless Communication

- 1) Draw and explain the complete block diagram of an OFDM transmitter and receiver. Explain the function of each block.
- 2) Explain the important parameters involved in selecting a modulation scheme for an OFDM system.
- 3) What is pulse shaping? Explain the need for pulse shaping in digital communication systems
- 4) Define spectral efficiency. Derive the expression for spectral efficiency of an OFDM system and explain the factors affecting it.
- 5) Explain timing synchronization and frequency synchronization in OFDM.
- 6) Explain channel assignment strategies in cellular systems.
- 7) Explain the relationship between frequency reuse, interference and capacity.
- 8) What is GPRS? Explain its architecture, features, and working in detail.
- 9) Draw and explain the complete GSM architecture.
- 10) Discuss the existing wireless communication technologies and their applications.
- 11) Explain the current trends in cellular communication systems.

MCQ based questions

The commonly used mathematical transform for OFDM generation is:

- A) FFT/IFFT
- B) ADC only
- C) DAC only
- D) AM

Answer: A

he cyclic prefix is inserted mainly to:

- A) Increase noise
- B) Reduce inter-symbol interference caused by multipath
- C) Reduce carrier frequency
- D) Increase modulation order

Answer: B

Which modulation is generally more robust against noise?

- A) 64-QAM
- B) 256-QAM

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- C) QPSK
- D) 1024-QAM

Answer: C

commonly used pulse-shaping filter is:

- A) Raised-cosine filter
- B) Low-pass DC filter
- C) Battery filter
- D) Optical filter

Answer: A

Windowing can improve the:

- A) Spectral characteristics of the transmitted signal
- B) Battery capacity
- C) Antenna height
- D) Number of bits in a symbol

Answer: A

Spectral efficiency indicates:

- A) How efficiently bandwidth is used to transmit information
- B) Transmitter physical size
- C) Antenna height
- D) Battery life

Answer: A

. Carrier Frequency Offset (CFO) can cause:

- A) Inter-carrier interference
- B) Increased battery capacity
- C) Reduced sampling
- D) Elimination of noise

Answer: A

The standard speech coding rate commonly associated with traditional GSM is approximately:

- A) 64 kbps
- B) 13 kbps
- C) 128 kbps
- D) 2 Mbps

Answer: B

BTS provides communication between:

- A) Mobile station and network
- B) HLR and VLR
- C) MSC and HLR only
- D) Two HLRs

Answer: A

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Which channel broadcasts system information to mobile stations?

- A) BCCH
- B) TCH
- C) SDCCH
- D) FACCH

Answer: A

A full-rate GSM traffic channel occupies:

- A) One time slot per frame
- B) Two time slots per frame
- C) Four time slots per frame
- D) Eight time slots per frame

Answer: A

FACCH is used when:

- A) Urgent signalling is required
- B) No signalling is required
- C) Only SMS is transmitted
- D) Only frequency correction is required

Answer: A

Which technique helps increase cellular network capacity by serving users with smaller coverage areas?

- A) Cell splitting
- B) Frequency elimination
- C) Modulation removal
- D) Trunking reduction

Answer: A) Cell splitting

The main objective of handoff is to:

- A) Maintain call/service continuity
- B) Reduce the number of cells
- C) Increase mobile size
- D) Stop communication

Answer: A) Maintain call/service continuity

The signal-to-interference ratio is commonly represented by:

- A) S/I
- B) I/S
- C) P/T
- D) R/N

Answer: A) S/I

Which technique can improve coverage in areas with weak signal strength?

- A) Repeaters
- B) Frequency elimination

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C) Reducing antenna height to zero

D) Removing base stations

Answer: A) Repeater

rade of Service (GoS) indicates:

A) Probability that a call is blocked or delayed

B) Battery quality

C) Antenna height

D) Data encryption level

Answer: A) Probability that a call is blocked or delayed